

EngineerVerse

Basic Electrical Engineering - Chapter 1: Introduction

What is Electrical Engineering?

It is the branch of engineering concerned with generation, transmission, distribution, control and utilization of electrical energy.

History

Important contributors: Franklin, Volta, Faraday, Edison and Tesla.

Applications

Power systems, EVs, renewable energy, automation, industries and electronics.

Basic Quantities

Current (I), Voltage (V), Resistance (R), Power $P=VI$, Energy $E=Pt$.

Safety

Use insulated tools, proper earthing, MCB/fuse protection and disconnect supply before maintenance.

Review Questions

Define current, voltage, resistance, power and explain the need for earthing.